

Ana Zahratul Firdausi

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PROFESSIONAL PROFILE

Informatics Engineering student from Brawijaya University with a focus on AI/ML Engineering and MLOps. Experienced in building and deploying end-to-end machine learning systems, including model training, experiment tracking, CI/CD automation, and production monitoring. Proficient in classification, clustering, and retrieval-based approaches using Random Forest, K-Means, and RAG pipelines (LangChain, FAISS), with hands-on experience in MLflow, DVC, Docker, and GitHub Actions. Currently gaining industry experience as a QA Intern and seeking an internship as an AI/ML Engineer to build reliable, production-ready ML systems.

EDUCATION

BRAWIJAYA UNIVERSITY
Undergraduate Informatics Engineering

Malang, Indonesia
2023 – Present

WORK EXPERIENCE

FACULTY OF COMPUTER SCIENCE - BRAWIJAYA UNIVERSITY

February 2025 - Juny 2025

Teaching Assistant for Object-Oriented Programming

- Assisted and led Object-Oriented Programming practical sessions for 29 students using a hands-on learning approach.
- Taught core OOP concepts including classes, objects, constructors, encapsulation, inheritance, polymorphism, as well as abstract classes and interfaces.
- Guided implementation of method overloading, final and static modifiers, arrays of objects, and collection frameworks.
- Supported students in applying exception handling and file handling for robust program execution and data management.
- Collaborated with faculty members to develop lab materials, assess student performance, and provide constructive academic feedback.

PT TELKOM INDONESIA - PADIUMKM

June 2026 - Present

QA Software Intern

- Wrote BDD Gherkin test scenarios for cancellation, order management, and seller notification flows within the Fulfillment Squad.
- Converted manual test cases into structured Gherkin scenarios covering Buyer Staff, Buyer Manager, and Seller role workflows.
- Debugged and investigated API issues, including 500 errors and authentication token failures, using browser Network tab analysis.
- Tested fulfillment-related workflows involving Tender Kilat, Denda, AWB, and tax document processes.
- Collaborated with the QA and development team to identify, document, and report bugs to support product reliability.

ORGANIZATIONAL EXPERIENCE

STUDENT EXECUTIVE BOARD OF COMPUTER SCIENCE FACULTY (BEM FILKOM UB)

January 2024 - January 2025

Expert Staff of Social and Environment Ministry

- Responsible for three work programs focused on the environment.
- Responsible for disseminating information related to environmental issues to the general public, especially FILKOM UB students.

RAION COMMUNITY .

April 2025 - December 2025

Product Manager

- Member of the Product Manager Apps division within RAION Community, a faculty-focused organization specializing in app and game development at Brawijaya University.
- Led strategic planning, execution, and management of diverse app projects, demonstrating robust project management skills, and contributed across the entire mobile application development lifecycle.

VOLUNTEERING EXPERIENCE

Chief executive of ENVIROLEARN

August 2024 – October 2024

- Planning and ensuring all event preparations run smoothly
- Dealing with external parties

Secretary of RECON 2024

July 2024 – October 2024

- Management of all required documents : make a proposal and LPJ, create all necessary letters.

Secretary of ESC 2024

Sept 2024 – Nov 2024

- Management of all required documents : make a proposal and LPJ, create all necessary letters.

Secretary of FILKOM Greeneration

May 2024 – July 2024

- Management of all required documents : make a proposal and LPJ, create all necessary letters.

Staff event division of PKKMB & SA FILKOM UB 2024

July 2024 – November 2024

- Person in charge of the PKKMB series
- Make a list of necessary equipment
- Create text for the Master of Ceremony
- Coordination with DDM division and equipment division
- LO of the speaker

Vice Chair of PKKMB & SA FILKOM UB 2025

April 2025 – December 2025

- Coordinated all committee divisions alongside the event chair.
- Served as main liaison between the committee and external parties (faculty, partners).
- Coordinated with lecturers and the faculty committee on event logistics.
- Oversaw event execution and stood in for the chair's operational decisions when needed.

PROJECT

FINAL PROJECT FOR THE ARTIFICIAL NEURAL NETWORK COURSE

As Machine Learning Engineer

- Designed and implemented the LVQ 2.1 and LVQ 1 algorithms manually without using existing machine learning libraries.
- Performed data preprocessing including Min-Max normalization, stratified train-test splitting, prototype initialization, and weight update rules according to the algorithm specifications.

EPICCSAFE - AI/ML CAPSTONE PROJECT

As AI/ML Engineer

- Designed, trained, and evaluated a Random Forest model for phishing and fraud website detection as the capstone project.
- Built and integrated a browser extension that flags phishing sites in real time using the trained model.
- Deployed the web application, accessible at epiccsafe.vercel.app, and prepared documentation and presentation materials for the project.

MLOPS FOR AIR QUALITY PREDICTION

As AI/ML Engineer

- Built a continuous training (CT) pipeline with three trigger types: performance degradation, data drift detection (Kolmogorov-Smirnov test), and scheduled retraining.
- Applied K-Means clustering (k=4) on PM2.5 thresholds to derive new air quality classification rules, and trained Random Forest models under both official (BMKG) and clustering-derived labels.
- Tracked experiments and model versions with MLflow, and managed datasets using DVC with a DagsHub remote.
- Diagnosed a critical generalization issue (high test accuracy but very low accuracy on unseen data) and resolved it using class-balanced weighting and stratified train-test splitting.
- Set up CI/CD automation for the training and deployment pipeline using GitHub Actions.

ACHIEVEMENTS

- Program Kreativitas Mahasiswa (PKM) - Futuristic Written Idea Division, Incentive | 2024
- Pagelaran Mahasiswa Nasional Bidang Teknologi Informasi dan Komunikasi (GEMASTIK) - ICT Business Development Division, Participants | 2024
- Pagelaran Mahasiswa Nasional Bidang Teknologi Informasi dan Komunikasi (GEMASTIK) - UX Research Division, Participants | 2025
- International Youth Excursion Network (IYEN) - SDGs Pitch Competition, 2nd Best Innovation Project | 2025
- Hology 8.0 - UI/UX Design Division, 2nd Runner-Up | 2025

SKILLS

- **Technical Skills:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, Machine Learning & Predictive Modeling, MLOps (MLflow, DVC, Docker, GitHub Actions CI/CD, Prometheus, Grafana), Retrieval-Augmented Generation (LangChain, FAISS, RAGAS), REST API Integration, Model Evaluation, Data Preprocessing, Microsoft Office Suite
- **Personal Skills:** Analytical Thinking, Problem-Solving, Critical Thinking, Communication Skills, Team Collaboration, Time Management, Adaptability, Leadership, Responsibility, Strategic Thinking, Learning Agility